

(2)

(b) Explain with example the scope and visibility of local static and global variables of the time of function execution.

Q.5 (a) Discuss advantage and disadvantages of recursion. Explain also the various types of recursion.

(b) Write a program in C using recursion to calculate factorial of a given number.

Q.6 (a) What do you mean by pointers? What kind of arithmetic operations can be performed on the pointers?

(b) Explain the following terms
(i) Command line arguments.
(ii) Enumerated data type and typedefs.

Q.7 (a) What is the difference of structure and union? How to create an structure in C language and assigned values to its member variables?

(b) Write a program in C language that read the contents of a file and print over the console.

Q.8 Write short notes on any two-

- (i) C preprocessor directives
- ii) Self referential structure
- (iii) Malloc () and calloc () function
- (iv) Testing and debugging.

[Total No. of Questions: 8]

[Total No. of Printed Papers : 2]

Roll No.....

MCA- 13

MCA– I Semester

Examination, Jan.- 2019

Programming and Problem Solving in “C”

Time: Three Hours

Maximum Marks: 70

Note: i) Attempt any five questions (each question carries equal Marks)

Q.1 (a) Explain various feature of a good programming.

(b) What is the meaning of an “algorithm”? Write an algorithm to find whether a give number is prime or not? Also draw the flow chart for the same.

Q.2 (a) Explain various important features of C programming language. Why it is considered as a middle level language?

(b) What are the importance of operator precedence and its associativity in order to solve an expression? Explain with suitable examples.

Q.3 (a) Differential the following with suitable example.

- (i) Type conversion Vs type casting.
- (ii) Static Vs register variables.

(b) What is an array data structure? Write a program that perform the transpose of two dimensional matrix .

Q.4 (a) Explain the various ways to pass parameters into the function in C language along with example.